

Literature Status: Main Metric X_p Conjectures Answered

Source arXiv:1408.5819; support arXiv:1601.03332

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Status. *Literature already answered.* The later paper arXiv:1601.03332 explicitly states that it resolves positively Conjectures 1.5, 1.8, and 1.12 from arXiv:1408.5819.

Source questions. Naor–Schechtman, arXiv:1408.5819, poses the following central metric X_p conjectures:

- Conjecture 1.5: for every $p > 2$, the metric X_p inequality for L_p should hold with the sharp scaling parameter $m \gtrsim_p \sqrt{n/k}$.
- Conjecture 1.8: for $2 < q < p$, if $(L_q, \|x - y\|_q^\theta)$ embeds bi-Lipschitzly into L_p , then $\theta \leq q/p$.
- Conjecture 1.12: for $2 < q < p$, the distortion of the grid $[m]_q^n$ into L_p should be asymptotic, up to constants depending only on p, q , to

$$\min \left\{ n^{\frac{(p-q)(q-2)}{q^2(p-2)}}, m^{1-\frac{2}{q}} \right\}.$$

Supporting answer. Naor, arXiv:1601.03332, states in the introduction that “Conjecture 1.5, Conjecture 1.8 and Conjecture 1.12” of arXiv:1408.5819 all have positive answers. The concrete results are:

- Theorem 1: for $p \geq 2$, every $f : \mathbb{Z}_{sm}^n \rightarrow L_p$ satisfies the metric X_p inequality when $m \geq \sqrt{n/k}$, giving the sharp scaling parameter.
- Theorem 2: for $2 < q < p$, the maximal snowflake exponent for L_q into L_p is exactly q/p .
- Theorem 3: for $2 \leq q < p$, the grid distortion satisfies

$$c_p([m]_q^n) \asymp_{p,q} \min \left\{ n^{\frac{(p-q)(q-2)}{q^2(p-2)}}, m^{1-\frac{2}{q}} \right\}.$$

Scope. This status packet clears those three main conjectures, not every open item in arXiv:1408.5819. In particular, arXiv:1601.03332 says that the convolution approach posed as Question 6.1 in arXiv:1408.5819 remains open.

References

- [1] Assaf Naor and Gideon Schechtman. *Metric X_p inequalities*. arXiv:1408.5819, 2014.
- [2] Assaf Naor. *Discrete Riesz transforms and sharp metric X_p inequalities*. arXiv:1601.03332, 2016.